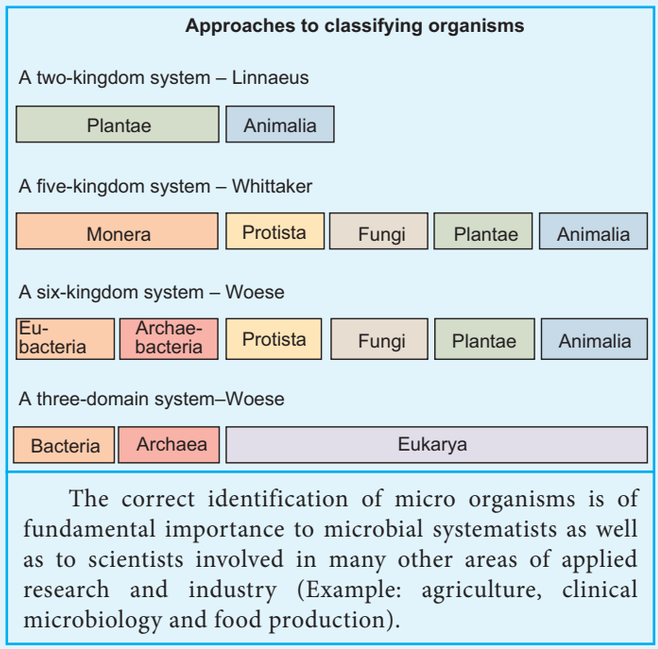
Chapter 8

Microbial Taxonomy

Chapter Outline

8.1 Diversity of Living Organisms is Fascinating
8.2 Binomial Nomenclature
8.3 Whittaker's System of Classification
8.4 Taxonomic Systems
8.5 The Three Domain System
8.6 The Past and Present Status of Bacterial Taxonomy



The correct identification of micro organisms is of fundamental importance to microbial systematists as well as to scientists involved in many other areas of applied research and industry (Example: agriculture, clinical microbiology and food production).

Learning Objectives

After studying this chapter the student will be able,

- *To understand the concept of taxonomy, taxon and phylogeny.*
- *To appreciate the contribution of Linnaeus and Whittaker.*
- *To learn the characteristics of Kingdom Monera, Protista, Fungi, Plantae and Animalia.*
- *To know some special methods used in classification of microorganisms.*

8.1 Diversity of Living Organisms is Fascinating

The branch of science which deals with the classification, nomenclature and identification of all living organisms is called Taxonomy. (Greek taxis means arrangements and nomos means law or to distribute). Because of large number and great diversity of organisms, biologists use the characteristics of different organisms to identify and group them. To understand life, it is essential to understand taxonomy. The method of grouping related organisms is the basis of classification (Figure 8.1). The objectives of taxonomy are:

- To establish the criteria for identifying organisms
- To arrange related organisms into groups
- To provide evolutionary information of the organism

The system of naming living organism is called Nomenclature.

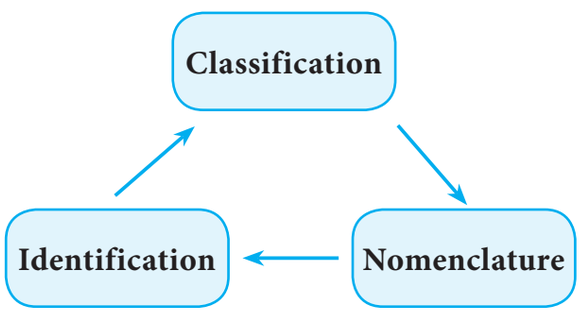


Figure 8.1: The three facets of taxonomy